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Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT After a load is restored or a line is reclosed, the remaining transient can resemble a new intervention. We use eight phasor measurement units (PMUs) and switched differential-algebraic predictors of the IEEE 39-bus system to compare a new event against the predicted recovery. A correct attribution names the event family and source before restoration. At a calibration target of one normal alarm per hour, the bank correctly attributed 99/108 interventions, compared with 72/108 for a continuous generalized likelihood ratio observer and 22/108 for ExtraTrees. The paired gain over the continuous observer was 25.0 percentage points (95% interval [13.89, 37.04] across 18 physical parents). The three procedures emitted 8, 85 and 0 unmatched alarms during event records, respectively. None emitted a normal alarm in 3.92 monitored hours per method. In a separate fixed-threshold control, resetting the estimated physical state reduced timely attribution from 108/108 to 75/108. These results quantify the coverage and alarm trade-off for sequential, nonoverlapping interventions from a known catalog in one simulator family. Independent per-PMU noise reduced load coverage.

INDEX TERMS change detection and isolation, differential-algebraic equations, event attribution, measurement integrity, phasor measurement units, physics-informed inference, power system dynamics.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, physical state continues to evolve. A second intervention may begin before that response decays. With eight PMUs, the observed change can combine recovery from the earlier event with forcing from a new source (Fig. 1(a)).

A detector may register this change without identifying its cause. We ask whether a monitor can name the new load or line before it is restored, and how many unmatched alarms it generates in doing so. Sequential detectors test for changes in the law of the observations [1]–[4]. Existing grid methods compare outage hypotheses or match transient trajectories to candidate faults [5]–[8]. Successive interventions create a further ambiguity because a restored asset can still account for part of the next measured transient.

We use switched differential-algebraic equation (DAE) models to predict each candidate response from the state inferred before its proposed onset. A restored history stops contributing to the weight assigned to an active intervention, but its predicted response can still explain incoming reports until the bank compresses it. The same update therefore supports three distinct outputs: an alarm, a source diagnosis and a restoration message (Section IV).

We evaluate the bank against a continuous generalized likelihood ratio (GLR) observer and an ExtraTrees classifier on held-out sequences with a known catalog. The GLR tracks one continuous state, and ExtraTrees scores sources from recent PMU features. Each procedure selects its threshold on calibration data under a target of one normal alarm per hour (Section VI). A separate reset experiment changes the estimated physical state within the bank and tests whether continuity matters. To establish where the result applies, we examine errors under network mismatch and independent per-PMU noise (Section VII).

II. RELATED WORK

A. CHANGE DETECTION AND ISOLATION IN DYNAMIC SYSTEMS

Sequential detection with a prior on the change time leads to the Shiryaev statistic, whose posterior-probability threshold is optimal for a geometric change-time prior [9]. Detection *and isolation* among several post-change laws was formalized

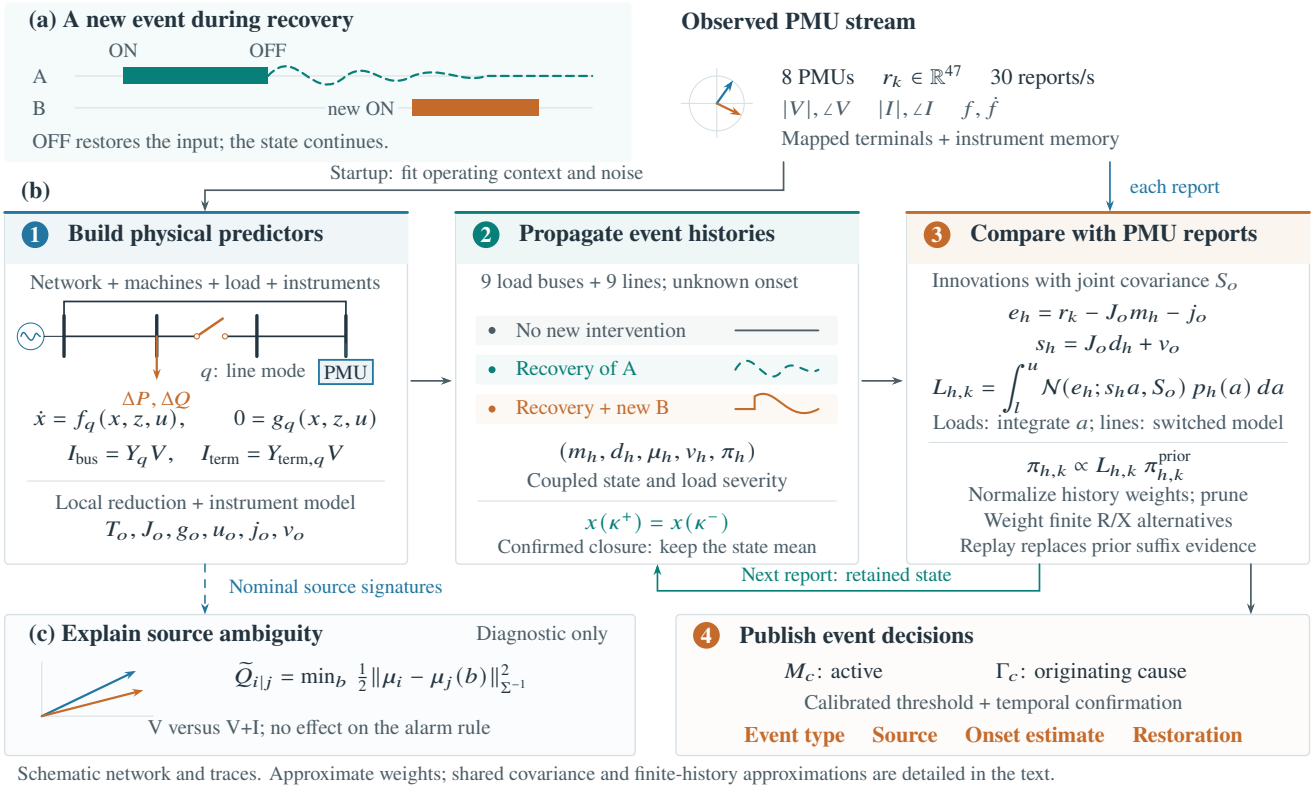


FIGURE 1. Recovery-aware event attribution from sparse PMUs. (a) A new intervention can begin while an earlier response persists. (b) Network-derived predictors compare recovery and new-event histories using the same report; integrated severity evidence updates their weights before detection, source diagnosis and restoration are published. The state continues through physical restoration; confirmed episode closure compresses histories to the retained state mean. (c) A separate nominal diagnostic measures source ambiguity; it does not gate publication. The electrical motif and traces are schematic.

by Nikiforov [2] and Lai [3]; asymptotically, the delay for a given error level is inversely proportional to the Kullback–Leibler information separating the true post-change law from its competitors [4]. For linear state-space systems, Willsky and Jones introduced signatures of jumps propagated through a Kalman filter and estimated their magnitude by generalized likelihood ratios [1]. Separating a constant bias from the state, as Friedland did for recursive filtering [10], gives the affine state–severity representation used in our load histories. Interacting and multiple-hypothesis filters propagate mode-conditioned states and mode-sequence histories with pruning and merging [11], [12]. Our method inherits these mechanisms. Its post-change models, however, are switched DAE operators of a power network, and a restored intervention remains a competing cause of later observations.

B. PHYSICAL INFERENCE OF GRID EVENTS

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [13]–[15]. Functional observability and descriptor-model diagnosability distinguish recovering a state quantity from identifying the input that produced it [16]–[19]. Phasor-angle hypotheses support outage identification [5], Bayesian outage inference with uncertain states [20], and quickest line-outage detection [6], [7]. Yang *et al.* add a time-varying AC likelihood chart [21] and particle-filter generator states [22] for persistent line outages, without restoration or loads. Anguluri *et al.* recover sparse forcing locations and inputs of linear dynamics by group LASSO [23], and sensor placement determines which sources are distinguishable [24]–[26]. Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window [8] and derive error/delay guarantees for an adaptive matrix CuSum [27]. Trajectory sensitivities and Gaussian evidence have been combined for parameter inference in power-system DAEs [28], [29]. PMU-based nodal-inertia estimation extends this neighboring line of work to spatially resolved dynamic parameters under ambient and transient conditions [30]. State and parameter estimates do not by themselves identify the intervention that produced them. Here the DAE-derived signatures provide a profiled separation diagnostic at a supplied onset, which we compare retrospectively with attribution outcomes; it is not a general detectability guarantee. Le *et al.* recently proposed a physics-informed graph neural operator that maps sparse, intermittent PMU windows to unmeasured rotor angle and speed trajectories, with swing equations in the decoder and a learned correction for model mismatch [31]. That preprint addresses dynamic state reconstruction. Our method identifies which intervention explains observations while an earlier response is still evolving.

C. LEARNED DIAGNOSIS FROM SPARSE MEASUREMENTS

Learned and knowledge-driven representations identify PMU disturbances and assets [32]–[34]. For sparse-PMU transient-stability assessment, Wang *et al.* combine dominant-bus selection with a bidirectional recurrent assessor and an online check of model applicability under system changes [35]. This is a useful precedent for monitoring whether a learned assessment remains valid, but its output is a stability assessment rather than a source attribution for successive physical events. Barati's gated-recurrent-unit model predicts disruptions from historical synchrophasor streams, a forecasting task distinct from assigning an observed intervention to a physical source during recovery [36]. SHRED reconstructs IEEE-39 dynamic states from a few PMUs [37], and Dey *et al.* localize distribution topology changes with hardware-in-the-loop validation [38]. Graph neural networks encode the grid topology in the learned map, for event identification [39], physics-informed state estimation [40] and localization after reconstructing unmeasured trajectories [41]. These applications span state reconstruction and event diagnosis. In the present attribution task, the physical bank derives candidate responses from the network equations. Its signature analysis describes separation under the assumed plant and noise model. ExtraTrees provides a learned comparison on the same 18-source catalog.

D. PMU MEASUREMENTS

The bank takes phasor reports as inputs. Synchrophasor estimation citeislam2026sogi,adetayo2026pllpmu, transient measurement tests citebecejac2019pmu, and PMU co-simulation for state estimation citeceleita2023iec61850 address how those inputs are produced or validated. Multisnapshot state estimation relates instrument errors to uncertain line parameters citedantona2026multisnapshot. For event attribution, the practical question is whether measurement errors hide or mimic a source response. We test independent per-PMU noise in the article and describe integrity checks in the supplement.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the reference record released with the SGSMA 2026 competition. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The causal attribution test addresses load changes and line openings with restoration; missing and corrupted data are diagnosed by the integrity relations of the supplementary integrity analysis. Faults and generation changes provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the definition of absolute frequency. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of $-120°$ and $+120°$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Figure 4 maps these physical and measurement operations to the report consumed by inference. Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is generated by the noise model described in Section VI.

An intervention history is the discrete tuple $h = (e, c, \tau, \kappa, \mathcal{I})$: family, source, onset, restoration time ($\kappa = \infty$ for a sustained event) and, for a load, a severity interval $\mathcal{I}$. The severity $a \in \mathcal{I}$ is a continuous latent variable conditioned on h and integrated in the evidence; a line opening has no severity. A load changes its local P/Q setpoints by the common factor $1 + a$; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \tag{4}$$

TABLE 1. Closest inference formulations and their relation to this study. The primary comparison uses a continuous GLR and ExtraTrees. Chen *et al.*'s detector is evaluated on persistent openings.

Route	Mechanism	Evaluated target	Role in this study
Willsky–Jones GLR; Nikiforov/Lai	Filter signatures; detection–isolation statistics	One jump or one change in a fixed system	Information analysis (Section V)
IMM / M3H	Mode-conditioned filters, history pruning and merging	Hybrid switching-state estimation	History-bank mechanism
Chen / Rovatsos	Network likelihoods, sequential tests	Line outages	Chen executed on persistent openings
Yang	Time-varying AC likelihood chart; particle-filter states with EWMA	Persistent line outages, ten PMUs	Closest dynamic precedent, not executed
Warner high-fidelity	Simulated V/I paths, onset search	Short-circuit line candidates	Closest trajectory precedent
DAE / stochastic-state estimation	Physical state and parameter models	States, inputs or parameters	Components
Wang et al. sparse-PMU TSA	Bus selection; recurrent assessor; validity check	Stability assessment under grid changes	Adjacent; no source attribution
Le et al. GNO (arXiv 2026)	Graph attention; swing-equation decoder	Machine-state trajectories from sparse windows	Adjacent state estimation; not event attribution
Learned PMU diagnosis	Features or learned representations	Event families and locations	Two ExtraTrees adapters executed
This work	Switched-DAE histories with recovery	Successive catalog load/line events	Attribution, alarm cost, nominal separation

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Parity diagnosis (supplement)
Unknown	Unresolved diagnosis	Not a new physical family

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after a 12-s warmup. This startup estimate assumes an intervention-free prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

The bank receives a stream of PMU reports and a catalog of load and line operators. Its first 300 reports establish a reference and initialize the filters. At each subsequent report, the bank propagates existing histories, proposes plausible recent onsets and updates each history's weight using the observed prediction error. A history records which asset changed and when its operators switched. A no-new-event history can carry the transient left by an earlier intervention. This prediction is the reference against which the bank evaluates a candidate new event (Fig. 1(b) and Algorithm 1).

A. NETWORK AND MEASUREMENT OPERATORS

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology q, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with a fixed angle reference, the derivatives at that consistent point

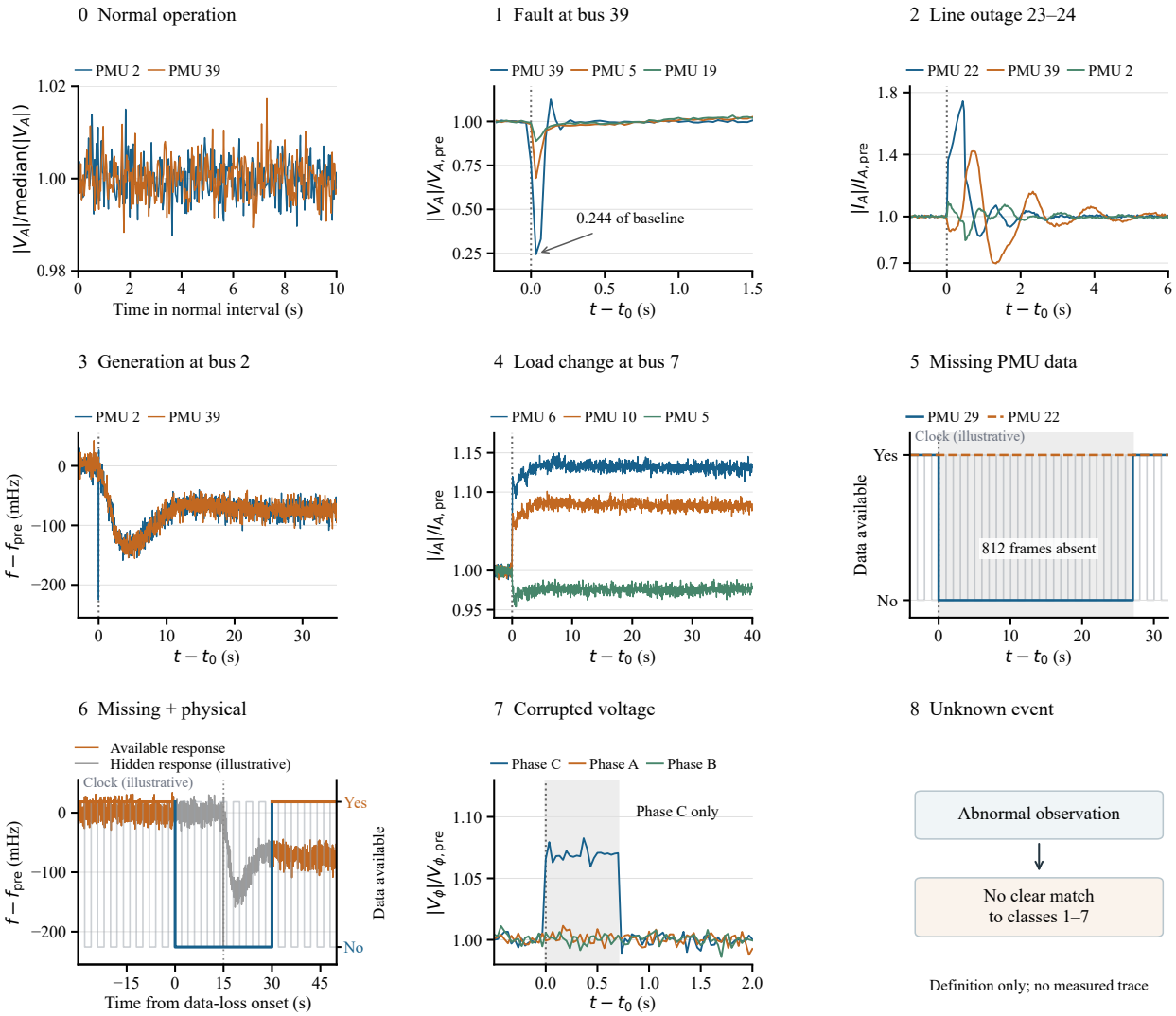


FIGURE 2. **Representative signatures from the archived RAW0001 reference simulation of the SGSMA 2026 competition, arranged by label 0–8. Class 0 shows a normal interval; classes 1–5 and 7 retain the original trace values. Class 6 applies an illustrative data-loss mask from 0 to 30 s to the PMU 2 frequency trace (generation onset at 15 s): orange shows the available response and Yes level; blue marks No, while gray shows the response hidden during the gap, not a detector reconstruction. The mask is not the recorded outage interval. The gray clock trains in panels 5 and 6 are illustrative, not measured report timing. Class 8 is a definition schematic, not an observed trace or a validated bank output. Event traces are aligned to source labels for illustration only. Only load and line mechanisms enter the main causal evaluation.**

give, suppressing ℓ,

$$\begin{aligned}\delta z &= -g_z^{-1}(g_x\delta x + g_u\delta u),\\ \delta\dot{x} &= c_q + A_q^{red}\delta x + B_q^{red}\delta u,\\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Branch stamps in g_q and H_q encode network connectivity; eliminating unmeasured algebraic variables generally produces dense reduced operators. A line switch changes both the voltage response and the terminal operator in H_q. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant. The supplement checks these derivatives against nonlinear trajectories and tests a relinearized physical predictor separately from the history bank.

Exponential discretization integrates c_q over each mode

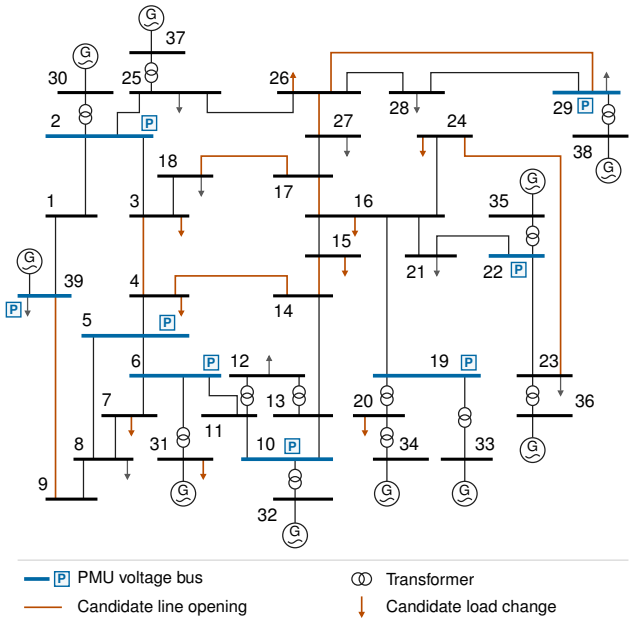


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract.**

interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining observation noise after instrument whitening. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section "Implemented Predictor and Conditional Uncertainty" gives the affine construction and its code correspondence. The network enters through these electrical operators, not a learned graph embedding. Updating conditional states and history weights does not adapt the physical parameters or measurement-noise model online. The supplement verifies a rotor-energy balance and its coupling through electrical graph ports. An invertible energy-coordinate representation preserves the physical predictions; energy reconstructed from the same PMU history is not scored as independent evidence.

B. STARTUP ESTIMATION

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. ONSET SEARCH AND HISTORY WEIGHTS

The method retains ten seconds of past conditional states. Its birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

As in separate-bias filtering [10], the conditional state mean is represented as $m_h + d_h a$; d_h is the Willsky–Jones signature of a unit severity propagated through the filter [1]. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h -$

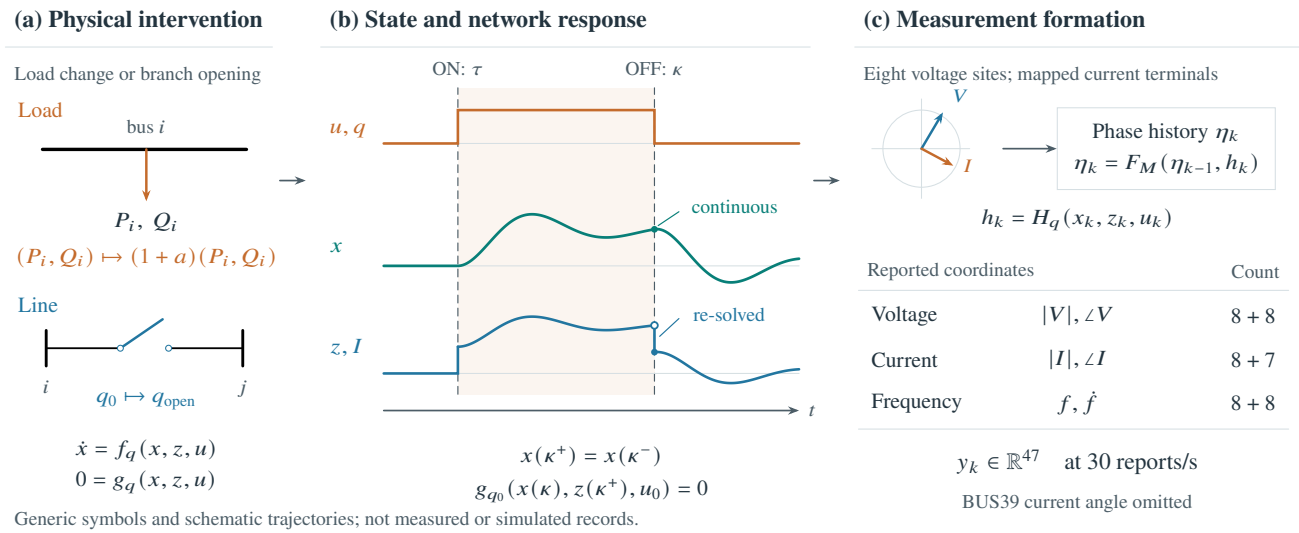


FIGURE 4. Illustrative chain from load/line intervention to causal PMU report. At OFF, the operator returns to baseline while x and phase-history η persist; algebraic voltages and currents are re-solved. Eight PMUs yield 47 coordinates at 30 reports/s; BUS39 current angle is omitted. Symbols and traces are schematic. Phase B/C are reconstructed from A; raw waveforms and harmonics are not inference inputs.

j_o and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$S_o = R + J_o P_\rho J_o^\top, \qquad K_o = T_o P_\rho J_o^\top S_o^{-1},$$
$$m_h^+ = T_o m_h + g_o + K_o e, \quad d_h^+ = T_o d_h + u_o - K_o s. \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter v_h. The affine conditional representation retains state–severity dependence:

$$\mathrm{Cov}(\xi, a \mid h) = d_h V_{a,h},$$
$$\mathrm{Var}(\xi \mid h) = P_\rho + d_h V_{a,h} d_h^\top. \tag{14}$$

Conditioning on past reports is implicit. The shared P_ρ solves the nominal-operator Riccati equation of regime ρ; S_o and K_o are then formed for each switched operator o. This is a steady-state approximation, not an exact covariance recursion along each switched history. At closure, the filter retains the augmented-state mixture mean and records the trace of the discarded amplitude and between-history covariance; it does not propagate that covariance. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. NETWORK MODEL VARIANTS

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight is updated with the one-step predictive density

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}, \tag{15}$$

$$p_\ell(y_k \mid \mathcal{F}_{k-1}) = \sum_{h'} \pi_{\ell,h',k-1} \sum_{h \in \mathcal{T}(h')} p_\ell(h \mid h')\, L_{\ell,h,k}. \tag{16}$$

The transition set $\mathcal{T}(h')$ contains continuation, restoration at either half-report phase for an eligible active history, and a birth of every catalog cell and severity interval at either phase from the no-intervention history; the transition probabilities sum to one, which the implementation asserts at every report. The density is evaluated before any state is corrected and before retrospective proposals are selected. Onsets proposed later for earlier reports change the inner history posterior through suffix replay but do not re-weight w_ℓ; this is the one point where the outer and inner recursions differ from a single joint posterior. Startup-fitting reports are not scored again. The mixture covers a finite set of network alternatives rather than a continuous impedance posterior.

E. STATE AND HISTORY AFTER RESTORATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\delta x_h(t) \simeq \Phi_0(t, \kappa)\delta x_h(\kappa),$$
$$\delta y_h^{\mathrm{net}}(t) \simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \tag{17}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

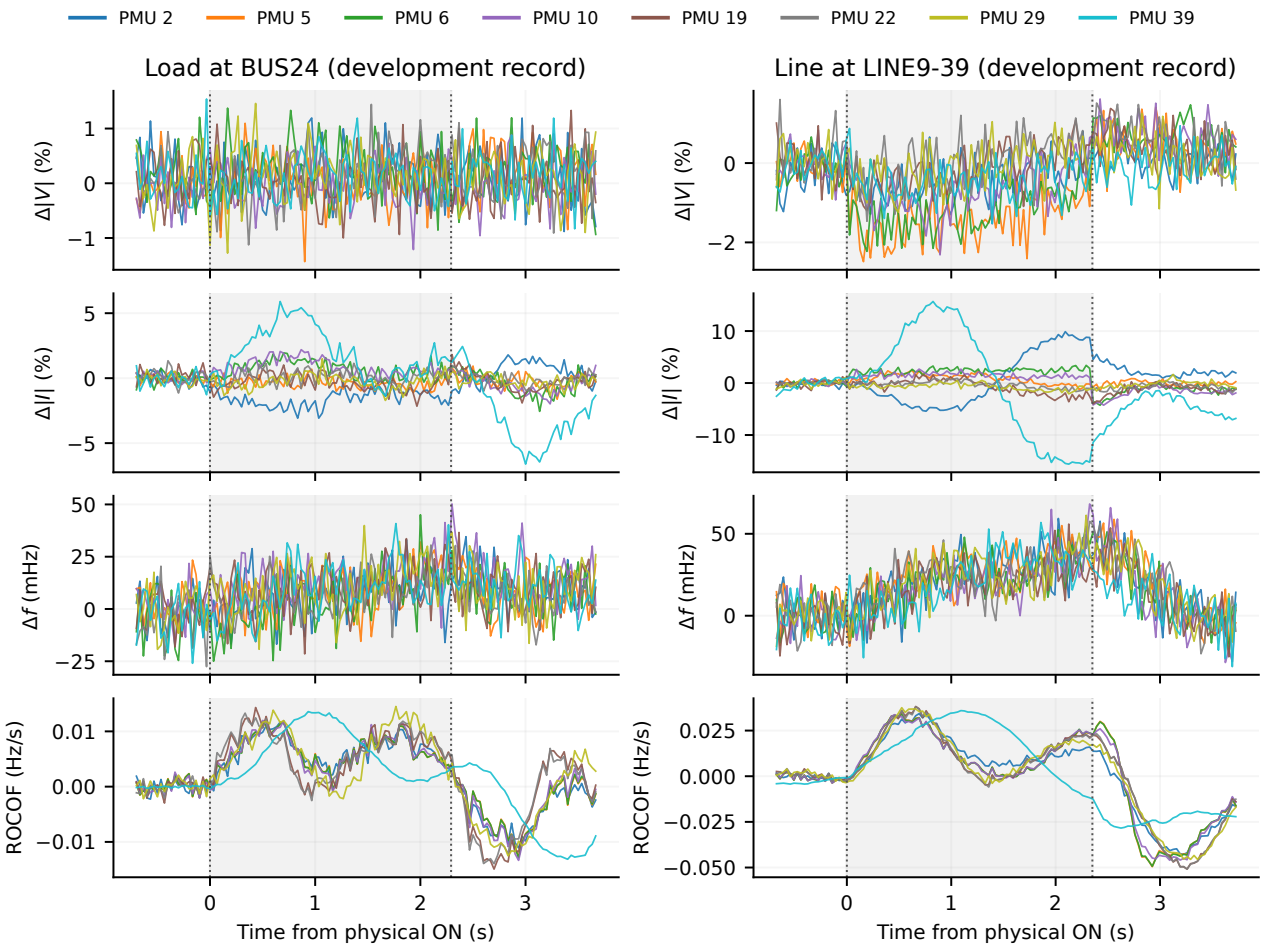


FIGURE 5. **Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.**

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$M_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\},$$
$$\Gamma_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \tag{18}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

The implementation uses M for active-event publication and Γ for source diagnosis. Their update weights are approximate because of pruning and shared steady-state covariance approximations; they are not calibrated posterior probabilities. A descriptive algebraic decomposition of ΔM into within-model evolution and network-model reweighting is given in the supplement; it does not enter the estimator's decision rule.

F. DETECTION, DIAGNOSIS AND RESTORATION RULES

An alarm is published when the active mass $\sum_c M_{c,k}$ exceeds 0.99 for six consecutive reports. The rule has the structure of a Shiryaev change-time threshold under the geometric prior [9], but pruning and covariance approximations make 0.99 an operational threshold on approximate mass, not a calibrated posterior-probability guarantee. Six-report confirmation suppresses brief excursions. OFF is published when the restoration mass exceeds 0.99 for 15 reports. The diagnosis reports the family with the largest cause mass and its highest-mass source; episode-replacement rules handle a different source that begins before the previous episode closes. Every message records its emission time and the last report used, so a diagnosis can be correct but too late to precede physical OFF.

These are the development publication settings. The common-budget comparison in Section VII-A selects thresholds separately on calibration records. The nominal bank uses

only the nominal network hypothesis.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$, fixed during method development before the independent schedules. They are deliberately conservative: test interventions last 2–5 s and are separated by 3–8 s, much shorter than the prior means of 60 and 30 s. A low birth hazard raises the evidence needed for an alarm by $-\log \pi_c$; Section V compares this margin with nominal signature energy. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The supplement maps each operation to its implementation and lists the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h') \pi_{\ell,h',k-1}. \tag{19}$$

This is the mode-sequence update of IMM and M3H filtering [11], [12]. Three features are specific to the present problem: the modes are oriented electrical intervention operators derived from the network DAE, histories carry a restoration time so that a restored intervention remains a competing cause, and the output separates active intervention from originating cause through (18). Retrospective births reuse a stored pre-onset state, and confirmed closure compresses histories into the retained augmented-state mean.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

No event truth is available to the algorithm. Retrospective replay changes the inner history weights but does not re-weight the outer model densities. The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. Decisions use only reports already received. Computation time is reported separately in Section VII.

Algorithm 1 Causal physical-history attribution

Require: Fixed plant/catalog, configuration, and PMU reports as they arrive
Ensure: Time-stamped detection, source diagnosis, and restoration messages

Fit operating context, reference, and noise from the permitted prefix
Initialize model/history states, weights, and pre-onset state store
for each newly received report y_k after startup **do**
    Map y_k to inference coordinates r_k
    Predict each retained history; carry physical state through OFF
    Save outer-model predictive densities before screening
    Propose eligible births from stored pre-onset states
    for each accepted proposal **do**
        Replay its suffix from the saved state, replacing old suffix scores
    end for
    Score histories before state correction; integrate load severity
    Apply line operators; correct states and update history weights
    Prune the beam; record discarded probability mass
    Update outer-model weights from the saved predictive densities
    Aggregate active-event mass M_k and cause mass Γ_k
    Apply consecutive-report alarm, diagnosis, and OFF rules
    if OFF is confirmed **then**
        Compress closed histories into the retained augmented-state mean
    end if
    Store current states and emit messages with the last report used
end for

V. NOMINAL SIGNATURES AND STATE CONTINUITY

The local operators give a nominal signature for each candidate. We use these signatures to explain separation and the effect of resetting the physical state under the assumed model. The numerical comparisons with completed runs are retrospective.

A. SIGNATURES AND INFORMATION

Let intervention i start at report 0 with severity a and let the plant follow the local switched model (8) without noise. Run the conditional filter of candidate history j, started at the same onset, on these reports. Its innovation $e_j(k)$ and severity sensitivity $s_j(k)$ follow the recursion (13). The filter of the true history at the true severity has zero deterministic innovation in this construction. We measure nominal separation by a

profiled signature energy,

$$\Lambda_{ij}(k) = \tfrac{1}{2} \min_{b\in[l,u]} \sum_{r=0}^{k} \left\| e_j(r) - s_j(r)\, b \right\|^2_{S_j^{-1}}, \qquad (20)$$

where a line candidate has no severity to minimize over. The quantity Λ_{i0} compares against the no-intervention path; $\min_{j\neq i} \Lambda_{ij}$ measures separation from the nearest alternative at the supplied onset. These are deterministic diagnostics. For fixed Gaussian paths with means m_i, m_j and joint report covariances Σ_i, Σ_j, the expected log likelihood ratio under i is

$$D_{ij} = \tfrac{1}{2}\big[(m_i - m_j)^\top \Sigma_j^{-1}(m_i - m_j) + \operatorname{tr}(\Sigma_j^{-1}\Sigma_i) - d + \log(|\Sigma_j|/|\Sigma_i|)\big], \qquad (21)$$

where d is the stacked report dimension. The covariance terms cancel when the covariances coincide. Equation (20) profiles severity on deterministic signatures using the candidate innovation covariance; it is not, in general, the full Gaussian KL divergence above. Optimizing a deterministic signature, profiling severity from noisy observations and integrating it under a prior are different operations.

For an idealized comparison of one onset and source against the null,

$$\lambda^\star = \log 99 - \log \pi_c. \qquad (22)$$

This is the score required for odds of 99 to 1 if π_c is used as the approximate prior odds. For a load with $\Lambda_{i0} = a^2 G_i/2$ and covariance scaled by α^2, the corresponding reference amplitude at horizon T is

$$a_{\mathrm{ref}}(\alpha, T) = \alpha\sqrt{2\lambda^\star / G_i(T)}. \qquad (23)$$

This does not specify a detection probability or a threshold guarantee for the bank: unknown onset, competing paths, marginal likelihoods and consecutive-report confirmation all affect publication.

B. THE COST OF FORGETTING THE PHYSICAL STATE

Consider a noise-free, matched-model continuation with no new intervention after an inferred closure at report k_c. Suppose the conditional mean at closure equals the augmented state. Carrying this state gives zero deterministic innovation. If the physical coordinates are set to zero while the instrument and discrepancy states are kept, the no-intervention filter receives the deterministic bias

$$\bar{e}_0(k) = C_0 \Phi_0(k, k_c)\, \delta x(k_c) - \sum_{r=k_c}^{k-1} C_0 \Phi_0(k, r+1) K_0\, \bar{e}_0(r), \qquad (24)$$

where $\delta x(k_c)$ is the zeroed physical part of the augmented state, $\Phi_0(k,r) = T_0^{k-r}$ and $C_0 = J_0$ are the no-intervention operators of (13), and K_0 is its gain. This is the free recovery response (17) seen through a filter that tries to absorb it. Any birth hypothesis j that explains part of this bias has a profiled energy gain $\Gamma_j^{sp}(k) = \Lambda_0^{reset}(k) - \Lambda_j^{reset}(k)$, where both energies use the same reports from k_c to k. A gain above $\lambda^\star$ crosses the idealized score reference; it does not guarantee a published alarm. Under the same construction with carry, the null energy is zero, so every alternative has gain $\Gamma_j^{sp,carry}(k) \leq 0$. The positive part of the best gain is therefore zero. Reset bias depends on the excited modes, their decay, and their projection onto the observed channels.

C. RETROSPECTIVE DIAGNOSTICS

We evaluated the signature and reset diagnostics in the nominal quiet covariance regime. The true-path deterministic residual was below 10^{-8} nats. Fig. 6 summarizes separation under those conditions.

For a 10% load step, the one-second nominal separation from the null ranges from 4603 to 6.9×10^6 nats; for lines it is at least 4.9×10^6 nats. These large margins are consistent with the short delays under matched conditions. The reference amplitude under uniform noise scaling is highest at bus 31. Neither diagnostic accounts for independent per-PMU noise or a shifted plant. The reset diagnostic (Section VII-B) evaluates the deterministic effect of discarding a previously estimated state.

A separate channel diagnostic holds source amplitudes and a common temporal covariance fixed. At 1 s, adding terminal currents to voltage increases the minimum pairwise separation by a median factor of 1.35 across the 18 sources (range 1.03–8.41). Allowing rival loads to fit severity over the admitted interval gives a median factor of 1.53. The supplement derives both diagnostics and shows how fitting severity can reduce source separation. Neither result measures attribution accuracy.

A separate development calculation allows uncertainty in the severity of the preceding load. Across 1 296 nominal contrasts, the nearest-source separation falls by up to 8.4%, with 0 changes of nearest rival. This calculation covers one recovery direction at a time. Its setup and results are in the supplement.

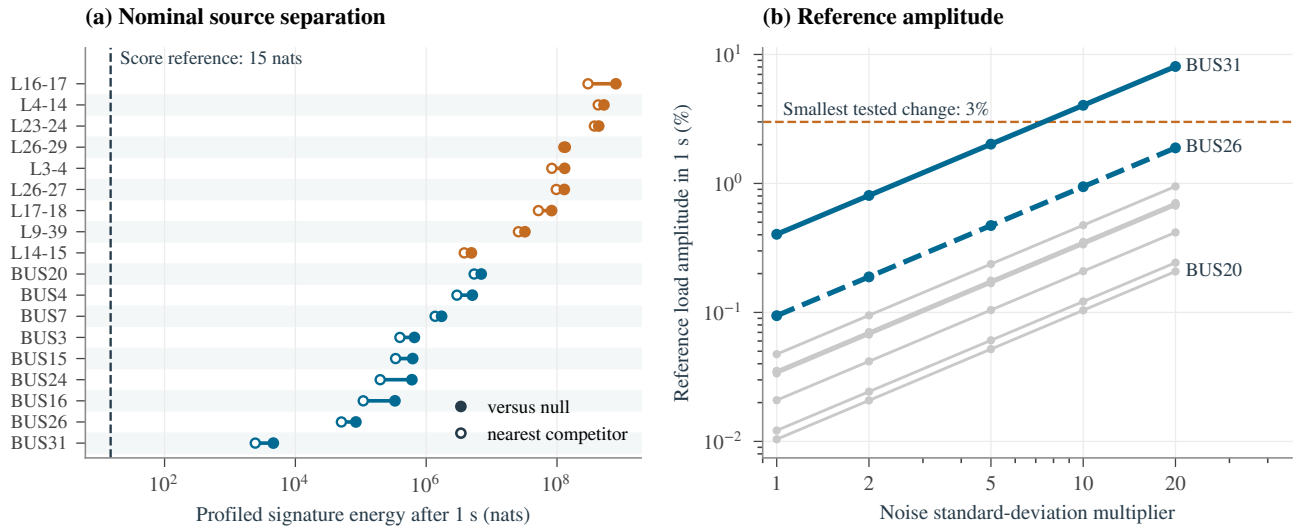


FIGURE 6. **Retrospective signature diagnostics using the nominal switched-DAE operators. (a) Profiled signature energy after 1 s against the no-intervention history (filled circles) and against the nearest competing source (open circles), for a 10% load step or a line opening; the dashed line marks the idealized score reference $\lambda^\star$. (b) Reference load amplitude at 1 s, (23), versus a uniform multiplier of the calibrated noise; gray lines are the other seven load buses.**

VI. EXPERIMENTAL DESIGN

A. PRIMARY COMPARISON

All three procedures see the same 30-report/s PMU stream and the same 18-source load and line catalog. The two physical procedures also share the nominal network and measurement operators and the allowed startup prefix.

We select each threshold using 12 event and 24 normal calibration records. A setting is feasible if it completes all records and its one-sided 95% Poisson upper bound is at most one normal alarm per monitored hour. Selection favors higher timely attribution, then fewer extra alarms, shorter diagnosis delay and a higher threshold, in that order. The GLR is rerun for each candidate threshold because publication changes its committed state. The held-out test has 18 event sequences with six interventions each and 24 normal records. The next intervention begins 3–8 s after OFF. We finalize the predictions before scoring them against reserved labels.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

We report unmatched alarms during event records separately from alarms during normal operation. The uncertainty analysis resamples whole physical parents in pairs (10,000 draws). Report timestamps measure decision delay. Wall-clock processing time measures computational cost.

B. PLANT AND SECONDARY TESTS

The plant is an IEEE-39 model in Julia/PowerDynamics with Sauer–Pai machines, IEEE Type I exciters, TGOV1 governors and constant-impedance ZIP loads. Each physical parent is integrated continuously, preserving differential states through ON/OFF. The PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001. At a switch-aligned report the export uses the post-switch algebraic value. The bus-31 candidate is the existing 9.2-MW/4.6-Mvar load at the generator bus that serves as slack only in the initial power flow; the dynamic model has no slack bus, so its load change is absorbed by machine dynamics like any other.

TABLE 3. **Procedures in the primary comparison. Each issues alarms and diagnoses through the common publication clock.**

Procedure	Evidence for a new source
History bank	Compare candidate intervention and recovery paths using their conditional states and integrated load severity.
Continuous GLR	Keep one committed augmented state through restoration. Compare candidate transitions over a 19-report search using a bounded profile fit for load severity.
ExtraTrees	Score sources from causal PMU features. Fit on disjoint simulation parents that cover the same catalog.

TABLE 4. **Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.**

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

Instrument noise follows a stochastic model fitted to normal segments of the competition reference record: per-family vector autoregressions with a strong common factor across PMUs. Voltage-magnitude noise is about 0.4% of nominal, but the noise of angle *differences* between PMUs is only 10^{-4} to 10^{-2} degrees, because most angle noise is common to all units. This small differential noise is what makes megawatt-scale changes observable at remote PMUs (Section V); it is a property of the reference data, smaller than the idiosyncratic error expected from field PMUs [42]. Section VII-D therefore tests larger and independent per-PMU noise.

The component study tests state continuity with the same observations, predictor and original thresholds in all arms. At inferred closure, one pair keeps the augmented state and the other zeros only the 114 physical state deviations. Within each pair, the load score either marginalizes or profiles severity. This is a retrospective comparison on previously examined records. It isolates the selected estimator changes at fixed thresholds.

An independent continuous evaluation varies network resistance and reactance (R/X) and instrumental noise across 18 event and 12 normal parents. It uses the original operating points of the five-model bank, the nominal bank and two earlier ExtraTrees adapters. Its persistent-line lane also includes the first-stop detector of Chen *et al.* [6]. The supplement records that lane and the comparator lineage. The independent per-PMU noise replay in Section VII-D changes the measurement law directly.

VII. RESULTS

A. TIMELY ATTRIBUTION AT A COMMON CALIBRATION BUDGET

The history bank correctly attributed 99/108 interventions before OFF, compared with 72/108 for the continuous GLR and 22/108 for ExtraTrees. The three procedures emitted 8, 85 and 0 unmatched alarms, respectively, during event records. The bank and GLR associated alarms with a similar number of interventions. The main difference was how often their diagnosis named the correct source before OFF.

Figure 7 shows a 25.0 percentage point gain over GLR (paired 95% CI [13.89, 37.04]), with fewer extra alarms. The gain over ExtraTrees is 71.3 points (95% CI [63.89, 78.70]), with more extra alarms. The intervals resample entire physical parents within the network/noise strata; events within a parent do not count as independent samples. All three methods had zero normal alarms in 3.92 monitored hours per method (one-sided 95% Poisson upper bound: 0.76/h). The result is conditional on the count model and the aggregate exposure. Candidate search, scoring and state commitment all contribute to the observed difference between the two physical procedures.

B. EFFECT OF PRESERVING THE PHYSICAL STATE

The nominal bank and the continuous GLR both carry physical state. Their primary comparison therefore cannot identify the effect of resetting it. In the fixed-threshold component study, resetting only the physical coordinates reduced timely attribution from 108/108 to 75/108 and increased unmatched alarms from 1 to 166 (Table 5 and Fig. 8). Marginal and profile severity scores gave the same outcomes within each state treatment. The simulation itself evolves continuously in all arms.

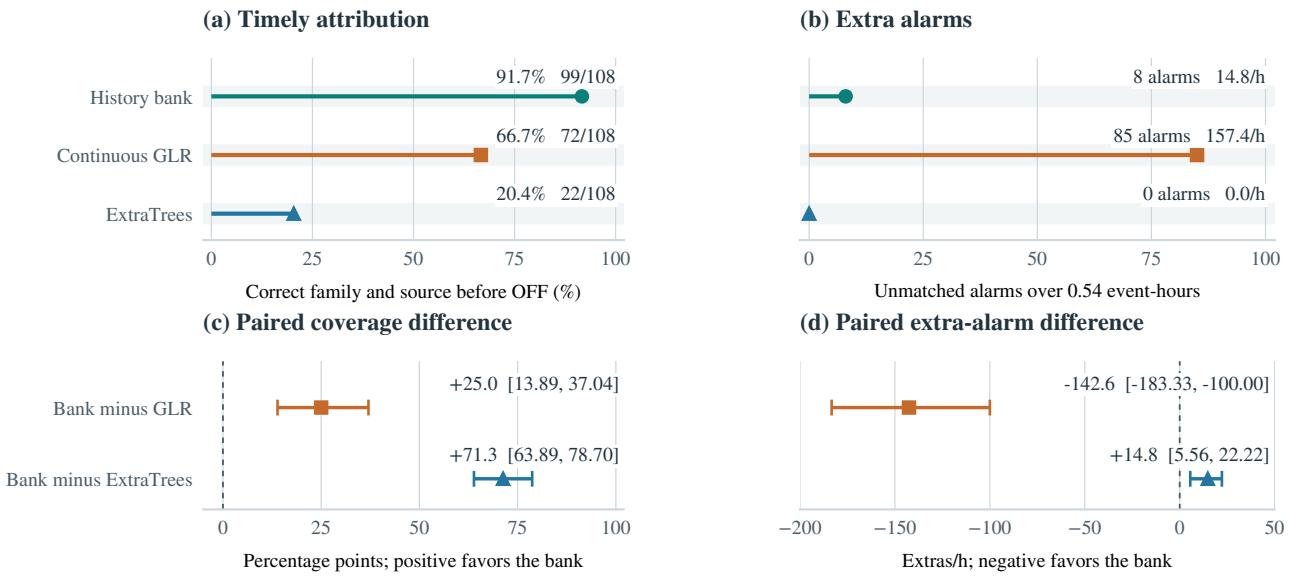
Fig. 9 compares this outcome with the retrospective reset diagnostic. The profiled energy gain exceeds $\lambda^\star$ after every intervention, by at least two and up to seven orders of magnitude, and the reset arm is harmed after 84 of the 108 interventions: an extra alarm in the following gap or a lost timely attribution of the next one. Harm occurs after 51/54 line events and 33/54 load events. The bus-31 events carry the smallest profiled gain and are harmed after three of seven. The diagnostic shows that reset leaves a residual that event candidates can explain. Its magnitude alone does not determine the observed outcome: episode handling and publication rules also matter. These are within-simulator contrasts at the original thresholds; the two scores are compared at equal thresholds, not at equal alarm rates.

C. SENSITIVITY TO NETWORK MISMATCH

The independent evaluation repeated 108 interventions in 18 parents under six network and noise conditions. The five-model bank correctly attributed 644/648 before OFF and the nominal bank 643/648. Of the five-model bank's 72 unmatched alarms, 68 occurred under the network perturbation outside the model set, as did its four late or omitted diagnoses. Neither physical bank alarmed in 11.8 monitored normal hours per method. The supplement gives the family breakdown, condition-specific counts and the comparisons at original operating points.

D. SENSITIVITY TO INDEPENDENT PMU NOISE

To examine sensitivity beyond the calibrated noise, we reran the nominal bank with the same equations and publication thresholds on four held-out calibration event records (12 load and 12 line interventions, matched network) and two 600-s normal records (0.33 h). Six noise scenarios were generated with the simulator's own noise stream: the original realization scaled by $\times 1$, $\times 5$, $\times 10$ and $\times 20$, and the original plus independent per-PMU relative-magnitude and angle noise, each with standard deviation $\sigma = 0.0005$ or 0.001 (angles in radians). The added perturbation has approximately 0.071% or 0.141% RMS total vector error (TVE), respectively, before combination with the original noise. Voltage-angle noise is propagated to frequency and ROCOF by the simulator's 21-report estimator. The $\times 1$ scenario reproduces the original predictions message for message. The bank re-estimates its noise scale and observation discrepancy from the noisy 300-report prefix; the operating point is kept at its nominal-noise estimate because derivative matrices exist only there. The implementation refuses to start when the estimated noise scale leaves its calibrated support of $[0.1, 10]$; for this test only, that bound was widened, and every other line of code is unchanged.



Normal records: 0 alarms in 3.92 h per method; one-sided 95% Poisson upper bound 0.76/h.
Paired 95% intervals: 10,000 resamples of 18 event parents. Each method meets the calibration alarm budget.

FIGURE 7. Coverage and alarm cost on held-out sequences. (a) Correct family and source before OFF, out of 108 interventions. (b) Unmatched alarms over 0.54 monitored event-hours. (c,d) History bank minus each comparator, with 95% intervals from 10,000 paired resamples of 18 event parents. Positive coverage differences and negative alarm-rate differences favor the bank. The bank improves both outcomes over GLR but trades more extras for higher coverage than ExtraTrees. Thresholds were selected on separate calibration data under a target of one normal alarm/h; normal exposure and its Poisson upper bound are shown separately.

TABLE 5. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

Scaling the calibrated noise by up to twenty preserves all line attributions in this auxiliary cohort. One load intervention, a 4.1% change at bus 26, is missed at $\times 5$ and $\times 10$ but not at $\times 20$. Its amplitude exceeds the nominal reference at all three levels. That disagreement illustrates why a signature-energy crossing cannot guarantee a published attribution. The median load alarm delay changes from 0.35 to 0.36 s. No scenario produces an extra alarm on event records or an alarm on the normal records.

Independent per-PMU noise has a larger effect: the bank attributes 3/12 load interventions at $\sigma = 0.0005$ and 0/12 at $\sigma = 0.001$, while the corresponding line counts are 12/12 and 10/12. A matched-signature diagnostic using only the added independent noise still assigns substantial energy to these load responses (supplement). That restricted diagnostic excludes source search and the bank's covariance approximation; it does not establish achievable attribution accuracy. The bank fits one noise scale per signal family while retaining the reference cross-PMU correlation. Independent noise raises the fitted angle-noise scale and weakens load evidence under that model. Estimating cross-PMU covariance is therefore a candidate remedy that requires a separate controlled evaluation. These results do not establish readiness for field PMUs.

TABLE 6. Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

E. COMPUTATIONAL COST

Table 6 records the stages of the evaluation campaign, which ran the five-model bank on one thread. Its history inference

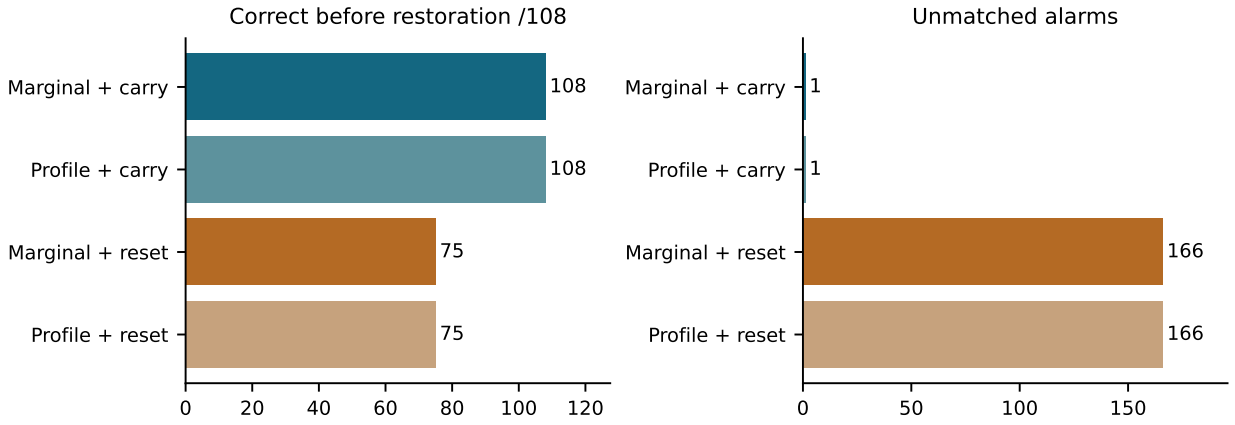


FIGURE 8. **Severity-scoring and state-carry study on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the alarm rate may differ between arms.**

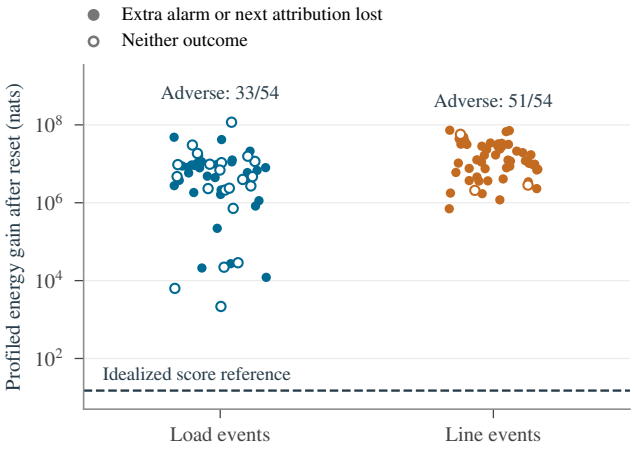


FIGURE 9. **Retrospective profiled energy gain after a state reset,** (24)**, for the 108 interventions of the controlled study. Filled: an extra alarm in the following gap or a lost timely attribution of the next intervention; open: neither. The dashed line is the idealized score reference, not the publication threshold. Under matched noise-free continuation, carrying the state gives no positive gain over the null.**

takes a median 358.7 s per 120-s record (99.6 ms per report), three times slower than the 30-report/s input, so the five-model bank as implemented is an offline causal replay. The cost is dominated by the number of network hypotheses. Re-running the unchanged nominal bank alone on the same hardware and thread settings (Section VII-D) gives 26.4 ms per report (median over 6 records, maximum 30.0 ms), below the 33.3-ms report interval; the nominal bank attributed 643/648 interventions in the independent evaluation. These are amortized processing rates on a desktop CPU; per-report latency percentiles and memory were not instrumented, and the 80-s startup bank construction is a one-time cost. Decision timestamps elsewhere in the paper are report times, not wall-clock latency.

VIII. DISCUSSION

A. WHAT THE COMPARISONS ESTABLISH

The reset experiment shows that discarding recovery state can turn genuine system motion into an apparent new event. The nominal signature analysis explains the mechanism: a reset leaves an innovation that a new-event candidate can fit. The size of this innovation alone does not fix the probability of an unmatched alarm because the bank still applies its proposal and publication rules.

The common-budget comparison asks a different question. The history bank reports more correct diagnoses before OFF than either comparator. Its unmatched alarm count falls between the GLR and ExtraTrees counts. The GLR retains a continuous state but also differs in onset search, score and commitment. The comparison thus measures the effect of the entire procedure. If the augmented state and transition model are sufficient for prediction, the full conditional state distribution captures the effects of prior histories. A mean alone can lose covariance and distinct alternatives. A controlled comparison that preserves the mean is needed to separate state carry from the benefit of retaining alternatives.

B. TRANSFER AND MEASUREMENT LIMITS

Observations and predictors use the same simulator family. The catalog is fixed, the gaps between interventions are 3–8 s, and no intervention overlaps another. Transfer of the dynamic bank to a distinct plant, shorter gaps and unseen sources are untested. Under independent per-PMU noise at 0.071% added RMS TVE, the bank misses most load changes. This limits transfer beyond the calibrated measurement model. The terminal map retains the provisional bus-5 current terminal and excludes the bus-39 current angle. A dropout with held observations prevents recovery in the auxiliary stress test. The bank did not start on the retrospective competition record; the integrity checks and separate attributor in the supplement do not validate the dynamic bank on that record.

The tested IEEE-39 plant also has no grid-forming inverter controller dynamics. Applying the method to converter-dominated networks would require adding the relevant con-

trol and current-limiting states and testing those predictors anew; GFMI surveys describe several such modeling choices [43].

C. APPROXIMATION AND PROCESSING COST

Pruning, steady-state covariances, startup fitting and compression make the attribution weights approximate. Retrospective proposals change inner history weights without updating the outer model weights. The nominal bank processed reports faster than they arrived in a measured replay on one thread; the five-model version did not. This timing is specific to the reported hardware and records.

IX. CONCLUSION

A restored load or line can leave a transient that resembles a new intervention. The proposed bank predicts that transient from the carried state and evaluates new source hypotheses against it. At a common calibration target, the bank correctly attributed 99/108 interventions before restoration, compared with 72/108 for a continuous GLR and 22/108 for ExtraTrees. During event records, the three procedures emitted 8, 85 and 0 unmatched alarms, respectively.

Resetting the physical state at inferred closure reduced coverage at fixed thresholds. The difference between the complete procedures leaves open how much explicit history retention adds beyond carrying state. The evidence covers nonoverlapping load and line events from a fixed catalog in one simulator family. Most unmatched alarms in the separate network-mismatch test occurred under perturbations outside the five-model set. Independent per-PMU noise reduced load attribution.

REFERENCES

[1] A. S. Willsky and H. L. Jones, "A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems," *IEEE Transactions on Automatic Control*, vol. 21, no. 1, pp. 108–112, 1976.

[2] I. V. Nikiforov, "A generalized change detection problem," *IEEE Transactions on Information Theory*, vol. 41, no. 1, pp. 171–187, 1995.

[3] T. L. Lai, "Sequential multiple hypothesis testing and efficient fault detection-isolation in stochastic systems," *IEEE Transactions on Information Theory*, vol. 46, no. 2, pp. 595–608, 2000.

[4] A. Tartakovsky, I. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Boca Raton, FL: CRC Press, 2014.

[5] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.

[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023.

[9] A. N. Shiryaev, "On optimum methods in quickest detection problems," *Theory of Probability and Its Applications*, vol. 8, no. 1, pp. 22–46, 1963.

[10] B. Friedland, "Treatment of bias in recursive filtering," *IEEE Transactions on Automatic Control*, vol. 14, no. 4, pp. 359–367, 1969.

[11] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.

[12] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.

[13] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[14] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[15] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[16] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.

[17] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.

[18] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[19] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.

[20] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 6, pp. 1140–1153, 2014.

[21] X. Yang, N. Chen, and C. Zhai, "A control chart approach to power system line outage detection under transient dynamics," *IEEE Transactions on Power Systems*, vol. 36, no. 1, pp. 127–135, 2021.

[22] ——, "Dynamic power systems line outage detection using particle filter and partially observed states," 2021, version 3.

[23] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.

[24] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.

[25] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.

[26] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.

[27] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.

[28] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.

[29] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," *Electric Power Systems Research*, vol. 263, p. 113688, 2027.

[30] D. Brahma, A. K. Singh, A. S. Mir, N. Senroy, and B. C. Pal, "An analytical method for nodal inertia estimation of power systems," *Journal of Modern Power Systems and Clean Energy*, vol. 14, no. 3, 2026.

[31] H. H. Le, H. Wang, C. Moya, M. Netto, and G. Lin, "Dynamic state estimation from sparse PMUs via physics-informed graph neural operators," 2026, arXiv:2611.10468, version 1, submitted November 7, 2026.

[32] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.

[33] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.

[34] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.

[35] G. Wang, J. Guo, S. Ma, X. Zhang, Q. Guo, S. Fan, and H. Xu, "Data-driven transient stability assessment using sparse PMU sampling and online self-check function," *CSEE Journal of Power and Energy Systems*, vol. 9, no. 3, pp. 910–920, 2023.
[36] M. Barati, "Faster than real-time prediction of disruptions in power grids using PMU: Gated recurrent unit approach," in *2019 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE, 2019, pp. 1–5.
[37] A. Pomarico, A. Berizzi, and J. N. Kutz, "A shallow recurrent decoder for dynamic state estimation with a limited number of PMUs in power systems," *arXiv preprint arXiv:2607.00116*, 2026.
[38] A. Dey, S. Chakraborty, and M. V. Salapaka, "Topology change estimation in power distribution networks: A machine learning approach," *IEEE Transactions on Smart Grid*, vol. 17, no. 4, pp. 3445–3462, 2026.
[39] Y. Yuan, Z. Wang, and Y. Wang, "Learning latent interactions for event classification via graph neural networks and PMU data," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 617–629, 2023.
[40] Q.-H. Ngo, B. L. H. Nguyen, T. V. Vu, J. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, p. 122602, 2024.
[41] M. Rimkus, P. Kokoszka, D. Duan, X. Wang, and H. Wang, "Graph neural networks for the localization of faults in a partially observed regional transmission system," *Scandinavian Journal of Statistics*, vol. 52, no. 2, pp. 572–594, 2025.
[42] IEC and IEEE, "Measuring relays and protection equipment—part 118-1: Synchrophasor for power systems—measurements," IEC/IEEE Std 60255-118-1:2018, 2018.
[43] D. B. Rathnayake, M. Akrami, C. Phurailatpam, S. P. Me, S. Hadavi, G. Jayasinghe, S. Zabihi, and B. Bahrani, "Grid forming inverter modeling, control, and applications," *IEEE Access*, vol. 9, pp. 114 781–114 807, 2021.

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